

Adaptations of the neuromuscular junction to exercise training

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The neuromuscular junction (NMJ) is the synapse that enables proper function of the neuromuscular system. Like other synapses, the NMJ is sensitive to changes in its pattern of use and adapts accordingly. Exercise training stimulates the NMJ and elicits functional and morphological remodeling resulting in improved performance. Physiologically, exercise training increases quantal content, safety margin, alters spontaneous release of neurotransmitter, and improves resistance to fatigue during a train of stimuli. Structurally, exercise training leads to enhanced presynaptic nerve terminal branching, number of vesicles, along with a greater number of postsynaptic receptors. Generally, presynaptic to postsynaptic coupling is unaffected by exercise. Although the NMJ responds to both endurance training (e.g. running, swimming) and resistance training, it appears that the constant stimulus of endurance training imparts a more powerful stimulus to the NMJ. The plasticity of the NMJ is elegantly demonstrated with exercise training.

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Introduction

The neuromuscular system is one of the most essential and adaptable of the 11 major physiological systems of the human body. At its core is the neuromuscular junction (NMJ) which is the synapse that joins the motor nervous system with the skeletal muscle fibers that contract enabling important and life sustaining functions including breathing, digestion, and locomotion. Moreover, the NMJ plays an essential role in the activities of daily living, as well as recreational activities such as sports that add to the quality of life.

As with all synapses, the NMJ demonstrates an impressive degree of plasticity in response to alterations in the

amount of activity that it experiences on a daily basis. For example, total (i.e. paralysis, denervation), and subtotal disuse (e.g. bed rest, limb immobilization) evoke detrimental adaptations such as alterations in the amount of the neurotransmitter acetylcholine (ACh) released both spontaneously, and when electrically stimulated [20,38]. Similarly, disuse results in significant presynaptic and postsynaptic morphological remodeling, including a greater number of nerve terminal branches, and increased dispersion of ACh receptors, respectively [2^{••},12,14]. Here, however, only adaptations to increased neuromuscular activity in the form of exercise training will be addressed.

Endurance exercise training

This form of increased neuromuscular activity is characterized by extended periods of continuous neuromuscular activity performed at a moderate, submaximal intensity. Commonly, this is presented in the form of running, swimming, or bicycling. Unfortunately, relative to inactivity, little research has been directly conducted into the effects of endurance training on the physiological function of the NMJ. However, in a particularly informative investigation, Fahim [17^{••}] examined electrophysiological parameters of neuromuscular synapses of the gastrocnemius muscles of adult rats who had completed a 12 week treadmill running program. When compared to untrained sex and age matched controls, it was determined that quantal content, or the amount of neurotransmitter released onto the postsynaptic endplate's ACh receptors with a single stimulus, was significantly increased among the trained muscles. Moreover, the safety margin of stimulation, or the degree of postsynaptic depolarization in excess of that required to open voltage gated postsynaptic ion channels, also was enhanced among trained NMJs. Each of these adaptations would act to ensure effective presynaptic to postsynaptic communication during prolonged periods of neuromuscular activity. And since whole muscle function is linked to effective NMJ function, these adaptations also postpone, and mitigate the onset of muscular fatigue during prolonged activity [2^{••}]. Similar training-induced improvements in NMJ transmitter release have been documented elsewhere [16].

One of the important enzymes that enables effective neuromuscular transmission is acetylcholinesterase (AChE). This enzyme is concentrated at the postsynaptic end plate in close proximity to ACh receptors [22,31,39]. AChE actually comprises several isoforms, each with its own specific function to play in neuromuscular transmission. While one specific isoform of AChE cleaves apart

the molecule of neurotransmitter bound to its receptor, another isoform is responsible for removing the byproducts of that reaction. In doing so, AChE clears the way for binding of a new ACh molecule to that receptor, and continued ionic flux across the postsynaptic membrane (see Figure 1). Several studies have reported that the specific isoform of AChE that is responsible for inactivating receptor-bound ACh, that is, G_4 , markedly increases its expression in the synapses of muscles endurance trained either via running [28], or swimming [24]. These adaptations improve protracted neuromuscular transmission, and by extension, resistance to muscle fatigue. Physiological adaptations of the NMJ to exercise can be found in Table 1.

Presynaptic morphological adaptations

In addition to physiological adaptations of the NMJ to endurance training, numerous morphological adaptations also have been observed. These include an increase in the number of presynaptic nerve terminal branches, enhanced total nerve terminal branch length and branching complexity, but a maintenance of average branch length of those same terminals [9,11,15,35,47,49^{*}]. While the absence of adaptation in average branch length may be surprising, it actually makes sense given the nature of the conduction of action potentials toward the active zones where docked vesicles containing ACh are positioned. The myelination along the axon that assists the movement of action potentials toward the nerve terminal

Figure 1

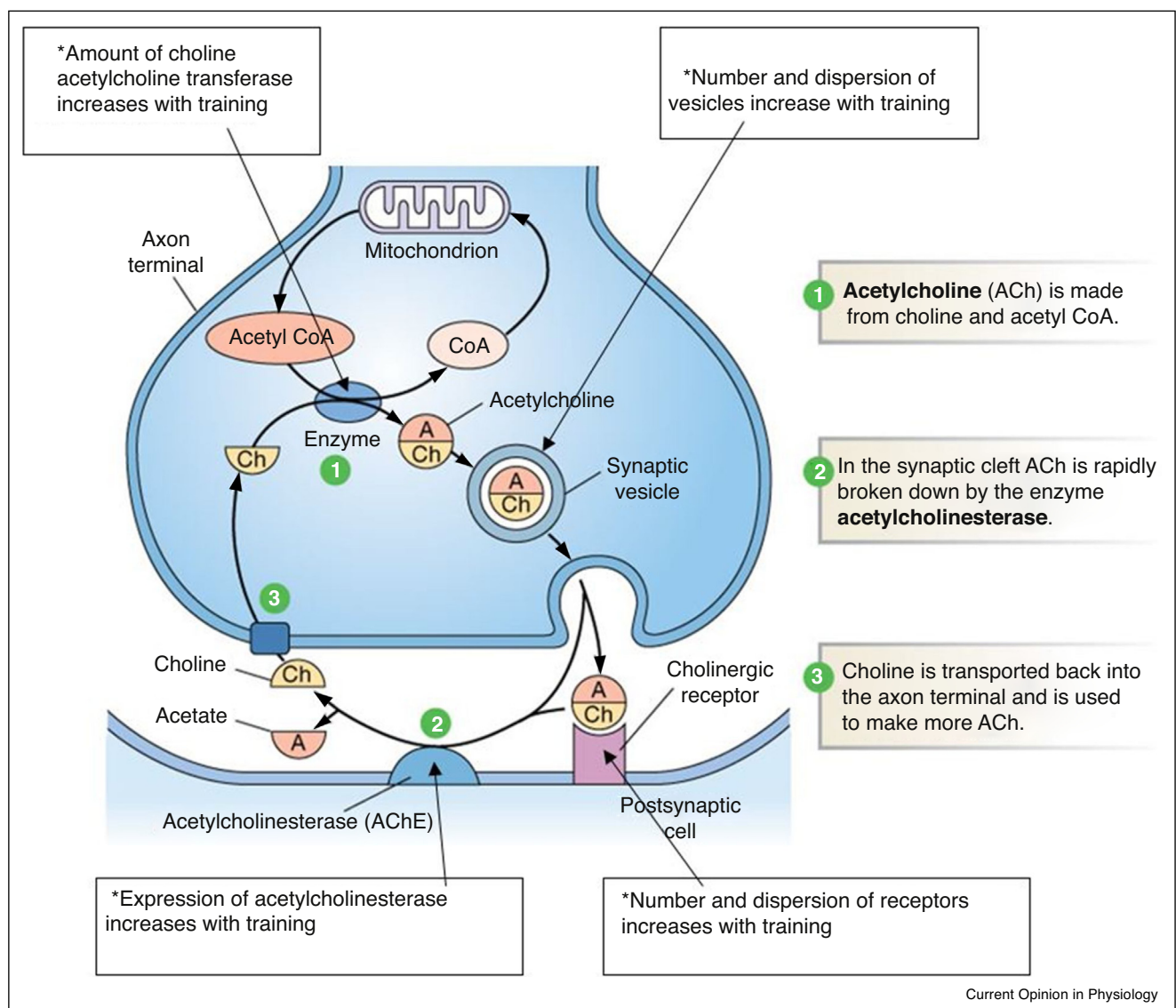


Illustration of how acetylcholinesterase (AChE) functions at the neuromuscular junction and examples of subsynaptic adaptations induced by exercise training.

Table 1**Exercise-induced physiological adaptations of the neuromuscular junction**

• Amplitude of mini-endplate potentials	↑
• Frequency of mini-endplate potentials	↔
• Amplitude of stimulated quantal content	↑
• Depression of quantal content during train of stimuli	↓
• Safety margin of neuromuscular transmission	↑
• Acetylcholinesterase activity	↑
• Choline acetyltransferase	↑

is no longer present at the axon's terminal endings. This is to expose the voltage gated calcium channels that allow an influx of extracellular calcium which releases vesicles docked at active zones so they can, via exocytosis, release ACh into the synaptic cleft. It is important to recognize that because of the absence of myelination at the nerve terminal endings, action potentials must rely solely on passive diffusion on their way to the active sites. Thus, nerve terminal endings cannot be too long otherwise a diminished depolarization amplitude would be inadequate to open the voltage gated calcium channels co-localized with docked vesicles. In that case, there would be no exocytosis of neurotransmitter, and by extension, failure neuromuscular transmission; this does occur under conditions of neuromuscular fatigue.

To counter this, endurance training increases the total length of presynaptic nerve terminal branching by augmenting that synapse's branch number, not its average branch length. Mechanisms fully explaining how exercise elicits increased terminal branching have yet to be fully elucidated, but it has been shown that neurotrophin 4 (NT-4) which stimulates nerve terminal branching, positively responds to exercise [21,36] thus enabling the NMJ to store ~30% more vesicles [10^{••},16]. It appears that training does not do this by increasing the density of ACh vesicles per given branch length, which might interfere with optimal vesicular recycling. Rather, total branch length at trained NMJs is increased [9]. Similarly, when examining the stained clusters of vesicles at endurance trained NMJs, it is evident that their distribution is more dispersed than what is noted at the nerve terminals of untrained NMJs [9]. Because of the ongoing recycling of vesicles during continuous stimulation [37,48,50], it is understood that this dispersion enables more effective recycling and endocytotic uptake of membrane following the release of ACh into the synaptic cleft. In essence, increased trafficking of vesicular membrane during continuous use occurring during endurance training requires

more space to enable efficient ongoing neuromuscular transmission.

The importance of the presynaptic active zone docking of ACh filled vesicles pending their release upon the influx of calcium has previously been described in full [26[•],33]. Recently, it has been found that subcellular modifications of the nerve terminal can be elicited by endurance training without accompanying cellular remodeling. More specifically, increases in the measured area of active zone bars, which anchor vesicles, have been detected as a result of endurance training without concomitant expansions of total nerve terminal length. Interestingly, it was noted in that same study that the ratio of vesicles to Bassoon – a major constituent protein of active zones – remains the same in trained and untrained NMJs suggesting that if more vesicles are to be located in trained NMJs, there needs to be a roughly equivalent increase in the number of docking sites at those synapses, as has been suggested elsewhere [32]. As another example of the sensitivity of NMJ remodeling, adaptations in presynaptic nerve terminals related to endurance training are confined to those myofibers, and NMJs, recruited during moderate intensity endurance training. Specifically, such remodeling is evident at low threshold, easily recruited slow-twitch myofibers, but not at higher threshold, less recruited fast-twitch myofibers and their NMJs [11,43]. Along with endurance training comes a greater rate of cellular protein turnover. In the motor neuron, newly synthesized proteins destined to function at the terminal endings must be transported from the cell body where protein synthesis occurs, to the nerve terminal arborizations via axoplasmic transport. A recent research project has confirmed that increased motor neuronal activity significantly enhances axonal transport capacity of that cell improving neuronal function of more active, that is, trained, NMJs [23].

Postsynaptic morphological adaptations

Along with presynaptic reconfiguration, postsynaptic remodeling of the NMJ occurs with endurance training. For example, increases in the number and dispersion of ACh receptors have been documented in endurance-trained NMJs [9,10^{••},15,25]. Interestingly, the training-induced expansion in the area occupied by ACh receptors matches that observed in the associated presynaptic area occupied by neurotransmitter filled vesicles (both ~30%). Indeed, it has been demonstrated that the presynaptic and postsynaptic features of the NMJ, particularly the ACh release and binding sites are tightly coupled, even in the face of the significant remodeling elicited by exercise training, or even aging [9]. In addition to the co-localization of presynaptic vesicles and postsynaptic receptors, presynaptic to postsynaptic coupling can also be determined by quantifying the area of postsynaptic binding sites relative to the length of presynaptic nerve terminal branching. When determined in this manner it is reaffirmed that synaptic coupling is maintained despite

Table 2**Exercise-induced morphological adaptations of the neuromuscular junction***Presynaptic*

• Number of nerve terminal branches	↑
• Total length of nerve terminal branching	↑
• Average length of nerve terminal branches	↔
• Complexity of nerve terminal branching	↑
• Quantity of acetylcholine vesicles	↑
• Quantity of acetylcholine receptors/branch length	↔
• Active zone size	↑
• Number of vesicles/active zone length	↔
• Presynaptic to postsynaptic coupling	↔

Postsynaptic

• Number of acetylcholine receptors	↑
• Dispersion of acetylcholine receptors	↑
• Total area occupied by acetylcholine receptors	↑
• Perimeter length encompassing acetylcholine receptors	↑

significant structural remodeling [12,13^{*}]. This stability of presynaptic to postsynaptic coupling is to be expected since unlike a complete, untraversed gap representing the synaptic cleft portrayed in many textbook images, presynaptic and postsynaptic factions are physically bound together by durable filamentous proteins such as fibronectin, laminin, and even calcium channels [34,40]. In short, despite considerable architectural alterations evoked by changes in neuromuscular activity, coupling between presynaptic release sites and postsynaptic binding sites are remarkably well maintained so that effective neurotransmission is assured. An excellent review of the importance of extracellular proteins, including fibronectins and laminins, in maintaining synaptic function despite overall remodeling has recently been published [18]. Morphological adaptations of the NMJ to endurance exercise are summarized in Table 2. Images of untrained (top) and trained NMJs (bottom) are presented in Figure 2.

Resistance exercise training

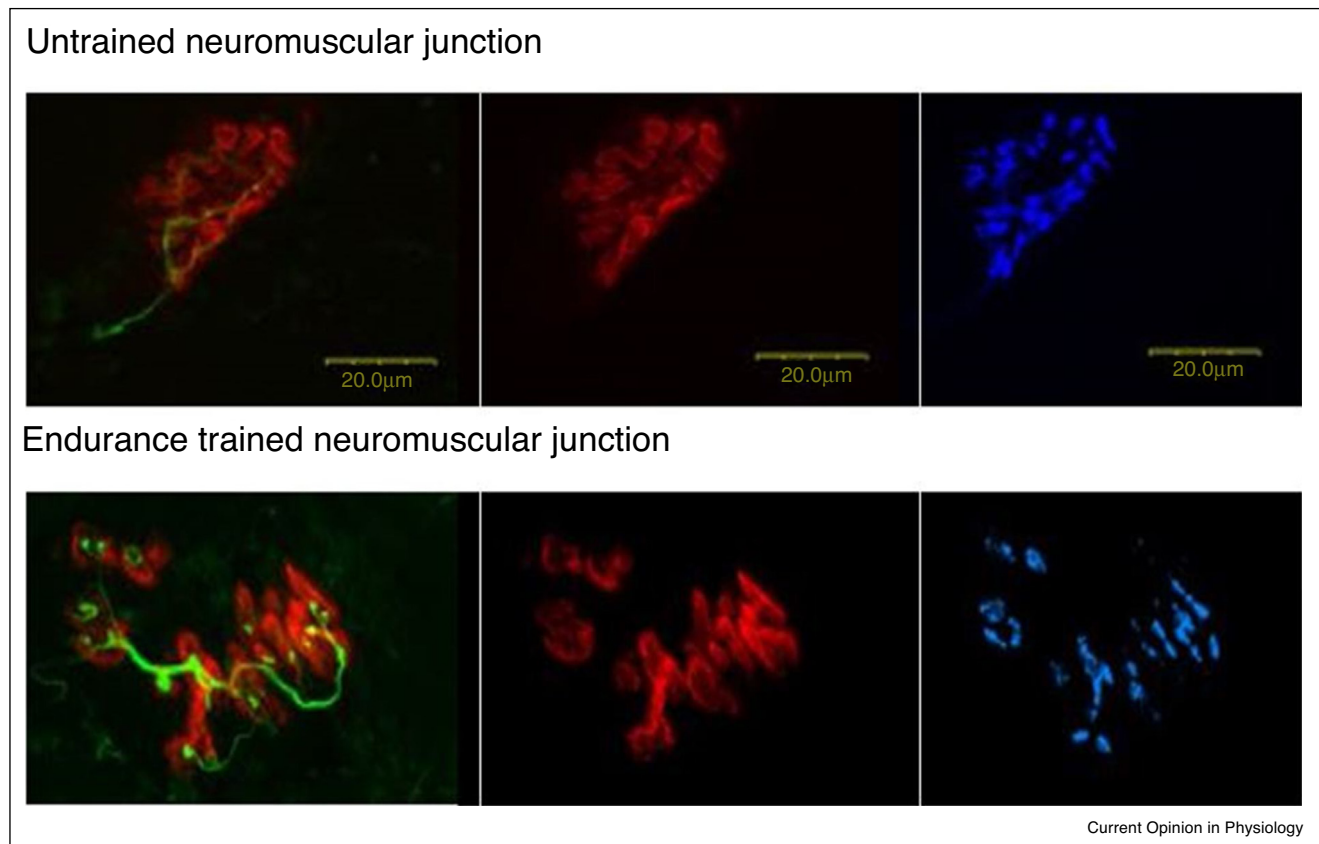
Another popular mode of exercise is resistance training, perhaps better known as weight lifting. While it is well known that resistance training elicits gains in muscle mass and strength [42], it is also a valued health maintenance intervention as it has been shown to improve the

symptoms of Type 2 diabetes, bone fragility, osteoporosis, and even hypertension [4,6,19]. Studying the effects of resistance training is difficult as small animals, with accompanying smaller muscles, are routinely used to examine the structural and functional characteristics of NMJs as they are located somewhere in the middle third of the whole muscle [41^{**},44]. Accordingly, the challenge is to establish a suitable model for weight lifting in rats, or mice [5,46]. Two commonly used models of resistance training for small animals is weighted ladder climbing, and synergist ablation to present constant overload to the unaffected synergistic muscle. While both of these methods have proven somewhat effective in stimulating muscle hypertrophy [1,7,27,30,45], neither faithfully mimics what is performed by a highly motivated human during resistance exercise trials. As a result, using such animal-based models to investigate the effects of resistance training must be approached with some degree of caution.

Nonetheless, there are some valuable insights suggesting that although weight lifting does result in some changes at the NMJ, those structural expansions are only about one half of what is reported as a result of endurance training, that is, ~15% versus ~30%, respectively. In addition, it appears that such NMJ adaptations to weighted ladder climbing are both muscle and myofiber type specific. Indeed, one study discovered that resistance training affected fast-twitch, but not slow-twitch myofibers by expanding the postsynaptic area occupied by ACh receptors, and even this was detected in soleus, but not plantaris muscles [13^{*}]. In short, resistance training was found to elicit postsynaptic NMJ enlargement, but only among the high threshold, lightly recruited fast-twitch myofibers that typically are only infrequently recruited and among the postural soleus, but not the non-postural plantaris muscle. Presumably, the exercise stimulus was intense enough to recruit those higher threshold motor units that typically are only rarely activated, thus eliciting NMJ remodeling. Yet in another investigation employing weighted ladder climbing as the exercise stimulus, the NMJs of the soleus muscle displayed significant size increments irrespective of myofiber type [8]. It is likely that the different results regarding the NMJs of the soleus shown by those studies [13^{*},8] are due to the longer training duration, and greater weight lifted while ladder climbing used in the study revealing significant NMJ adaptations to resistance training [8].

When evaluating the effects of chronic overload, imparted via synergist ablation, on NMJ structure, it was revealed that although this mode of resistance exercise was successful in producing significant (18%) hypertrophy of the plantaris muscle fibers, no accompanying restructuring of the NMJs in those muscles could be found [14]. In seeking answers to these disparate results among the two modes of resistance training, it is plausible that the constant contractile state of synergist ablation

Figure 2



Untrained (top) and endurance trained (bottom) neuromuscular junctions. In both panels, left image is of nerve terminal branching (green) with postsynaptic acetylcholine receptors (red), while the middle image is of postsynaptic receptors only (red), and the right image is of presynaptic acetylcholine vesicles (blue). Note that the trained synapse is larger with greater branching and more dispersed vesicles and receptors (scale bar = 20 μ m).

may trigger responses in the contractile, but not the excitatory proteins comprising the NMJ.

Natural postnatal development versus exercise

It has been previously shown that increases in the size of myofibers that naturally occur during postnatal development are associated with similar growth in NMJ dimensions [3]. But this relationship does not hold with modifications due to exercise training. For example, the myofiber hypertrophy resulting from chronic overload is not accompanied by similar NMJ growth [14]. Moreover, the moderate (10–15%) myofiber atrophy observed in endurance trained muscles is actually linked to significant enhancement of NMJ size [11,12]. These examples are clear evidence that when elicited by exercise, rather than progressive development, alterations in the morphology of NMJs do not necessarily emulate those of the myofibers on which they reside.

Conclusion

The NMJ is a unique synapse in that it is easily accessible, is simple in its connectivities, that is, a single pre-synaptic and a single postsynaptic cell, and displaying a normal postsynaptic depolarization from only a single pulse that exceeds the safety factor needed to generate a postsynaptic excitatory response by several-fold. None of these characteristics are exhibited by synapses of the central nervous system [29,41^{••},44]. Yet at the same time, the NMJ is similar to other synapses in that it demonstrates a high degree of plasticity, both physiological and morphological, in response to changes in its pattern of habitual use. Accordingly, this synapse demonstrates robust adaptability to both endurance, and resistance types of exercise training.

Future directions

Although much has been learned regarding the adaptations of the NMJ to exercise training over the last 30–40 years, this topic remains ripe with opportunity for

additional research endeavor. As previously mentioned, there is still much to be learned about the effects of resistance training for want of an adequate small animal model of weight lifting. Importantly, the focus of investigation of the NMJ and exercise science research should now turn to the subcellular, rather than the cellular level. Such subcellular topics might include adaptations of active zones, enzymes producing vital components such as choline acetyltransferase, internal vesicles, and proteins that properly dock those vesicles at active zones, among others. Information resulting from these studies would obviously result in a more complete understanding of how exercise training remodels the NMJ both physiologically and morphologically.

Conflict of interest statement

Nothing declared.

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